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Problem statement : Unemployment analysis with Python

Unemployment is measured by the unemployment rate which is the number of people who are unemployed as a percentage of the total labour force. We have seen a sharp increase in the unemployment rate during Covid-19, so analyzing the unemployment rate can be a good data science project.

Import Libraries

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Load Unemployment Dataset(CSV)

```
In [4]: df = pd.read_csv("Unemployment_Rate_upto_11_2020.csv")
```

View Dataset

```
In [5]: df.head()
```

Out[5]:

	Region	Date	Frequency	Estimated Unemployment Rate (%)	Estimated Employed	Estimated Labour Participation Rate (%)	Region.1	longitude
0	Andhra Pradesh	31-01-2020	M	5.48	16635535	41.02	South	15.9129
1	Andhra Pradesh	29-02-2020	M	5.83	16545652	40.90	South	15.9129
2	Andhra Pradesh	31-03-2020	M	5.79	15881197	39.18	South	15.9129
3	Andhra Pradesh	30-04-2020	M	20.51	11336911	33.10	South	15.9129
4	Andhra Pradesh	31-05-2020	M	17.43	12988845	36.46	South	15.9129

In [6]: df.tail()

Out[6]:

	Region	Date	Frequency	Estimated Unemployment Rate (%)	Estimated Employed	Estimated Labour Participation Rate (%)	Region.1	longitu
262	West Bengal	30-06-2020	M	7.29	30726310	40.39	East	22.98
263	West Bengal	31-07-2020	M	6.83	35372506	46.17	East	22.98
264	West Bengal	31-08-2020	M	14.87	33298644	47.48	East	22.98
265	West Bengal	30-09-2020	M	9.35	35707239	47.73	East	22.98
266	West Bengal	31-10-2020	M	9.98	33962549	45.63	East	22.98

Dataset Information

In [7]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 267 entries, 0 to 266
Data columns (total 9 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Region                                267 non-null    object
1   Date                                  267 non-null    object
2   Frequency                             267 non-null    object
3   Estimated Unemployment Rate (%)        267 non-null    float64
4   Estimated Employed                     267 non-null    int64
5   Estimated Labour Participation Rate (%) 267 non-null    float64
6   Region.1                              267 non-null    object
7   longitude                             267 non-null    float64
8   latitude                              267 non-null    float64
dtypes: float64(4), int64(1), object(4)
memory usage: 18.9+ KB
```

In [8]: `df.describe()`

Out[8]:

	Estimated Unemployment Rate (%)	Estimated Employed	Estimated Labour Participation Rate (%)	longitude	latitude
count	267.000000	2.670000e+02	267.000000	267.000000	267.000000
mean	12.236929	1.396211e+07	41.681573	22.826048	80.532425
std	10.803283	1.336632e+07	7.845419	6.270731	5.831738
min	0.500000	1.175420e+05	16.770000	10.850500	71.192400
25%	4.845000	2.838930e+06	37.265000	18.112400	76.085600
50%	9.650000	9.732417e+06	40.390000	23.610200	79.019300
75%	16.755000	2.187869e+07	44.055000	27.278400	85.279900
max	75.850000	5.943376e+07	69.690000	33.778200	92.937600

Check Missing Values

In [9]: `df.isnull().sum()`

```
Out[9]: Region      0
        Date        0
        Frequency    0
        Estimated Unemployment Rate (%)  0
        Estimated Employed  0
        Estimated Labour Participation Rate (%)  0
        Region.1      0
        longitude     0
        latitude      0
        dtype: int64
```

Data Analysis

```
In [10]: df.columns
```

```
Out[10]: Index(['Region', ' Date', ' Frequency', ' Estimated Unemployment Rate (%)',
               ' Estimated Employed', ' Estimated Labour Participation Rate (%)',
               'Region.1', 'longitude', 'latitude'],
              dtype='object')
```

```
In [11]: # strip leading and trailing spaces from all column names
df.columns = df.columns.str.strip()

# Verify the fix - you should see no extra spaces now
print(df.columns.tolist())
```

```
['Region', 'Date', 'Frequency', 'Estimated Unemployment Rate (%)', 'Estimated Employ
ed', 'Estimated Labour Participation Rate (%)', 'Region.1', 'longitude', 'latitude']
```

```
In [12]: # Average Unemployment Rate by Region(state)
avg_unemployment = df.groupby('Region')['Estimated Unemployment Rate (%)'].mean().s
print(avg_unemployment)

# Total Employment by Area (Region.1 corresponds to North,south,etc.)
region_employment = df.groupby('Region.1')['Estimated Employed'].sum()
print(region_employment)
```

Region	
Haryana	27.477000
Tripura	25.055000
Jharkhand	19.539000
Bihar	19.471000
Delhi	18.414000
Puducherry	17.942000
Jammu & Kashmir	16.477778
Himachal Pradesh	16.065000
Rajasthan	15.868000
Tamil Nadu	12.187000
Goa	12.167000
Punjab	11.981000
Uttarakhand	11.156000
West Bengal	10.192000
Sikkim	9.792500
Uttar Pradesh	9.737000
Kerala	9.434000
Andhra Pradesh	8.664000
Maharashtra	7.979000
Chhattisgarh	7.819000
Karnataka	7.668000
Madhya Pradesh	6.854000
Telangana	6.833000
Odisha	6.462000
Gujarat	6.376000
Assam	4.856000
Meghalaya	3.866000

Name: Estimated Unemployment Rate (%), dtype: float64

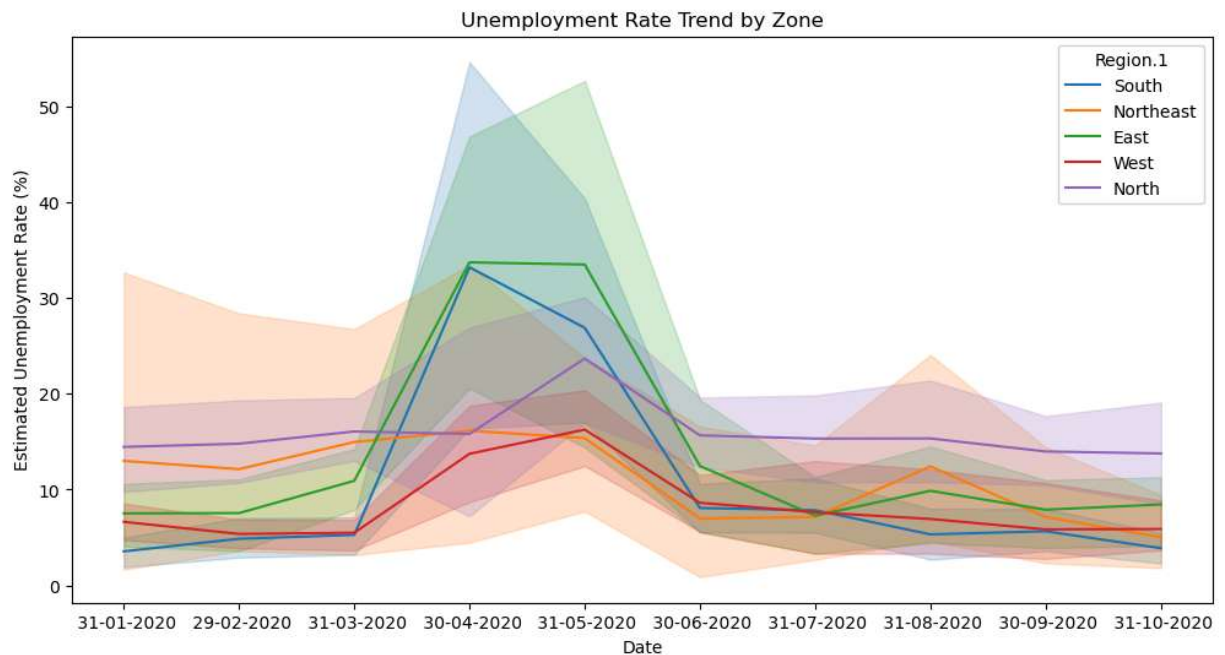
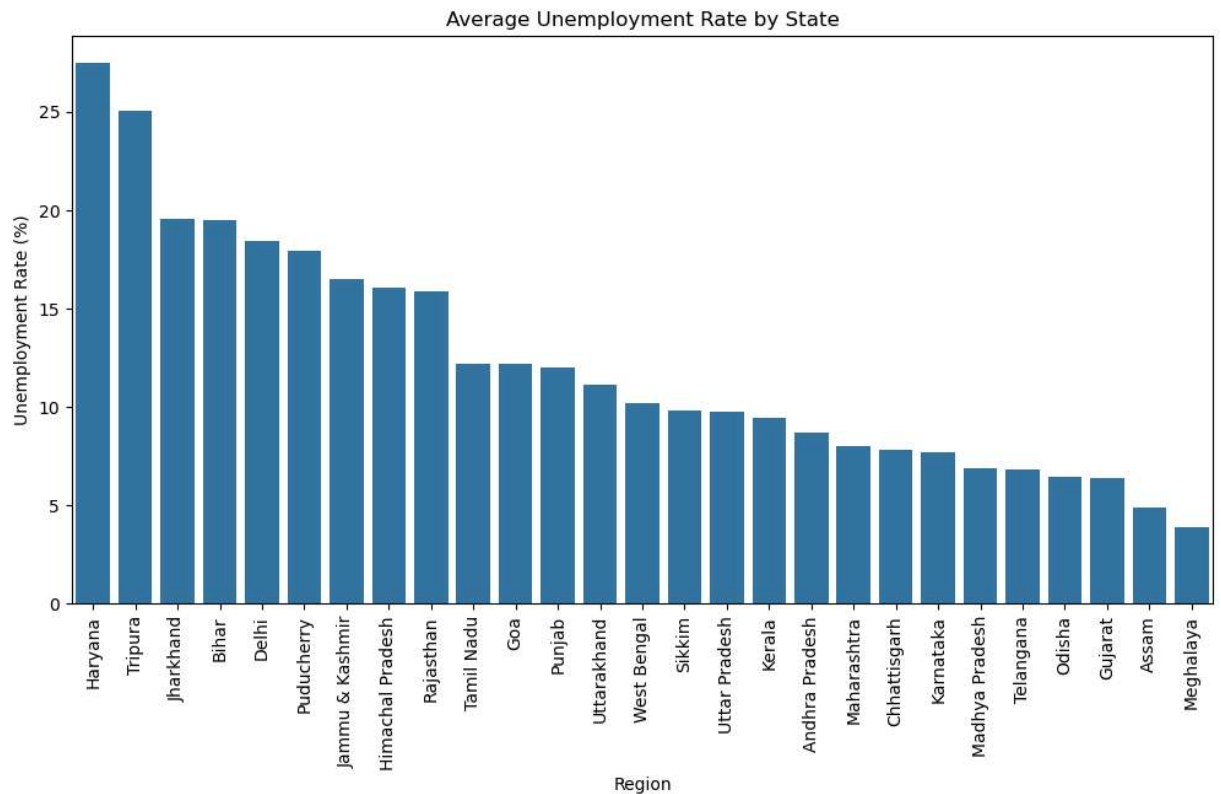
Region.1	
East	784094676
North	1032726546
Northeast	137450010
South	842435360
West	931175636

Name: Estimated Employed, dtype: int64

Visualizing the Data

```
In [16]: # Plotting Average Unemployment Rate by State
plt.figure(figsize=(12,6))
sns.barplot(x=avg_unemployment.index,y=avg_unemployment.values)
plt.xticks(rotation=90)
plt.title('Average Unemployment Rate by State')
plt.ylabel('Unemployment Rate (%)')
plt.show()

# Time series plot of unemployment rate over months
plt.figure(figsize=(12,6))
sns.lineplot(data=df, x='Date', y='Estimated Unemployment Rate (%)', hue='Region.1')
plt.title('Unemployment Rate Trend by Zone')
plt.show()
```

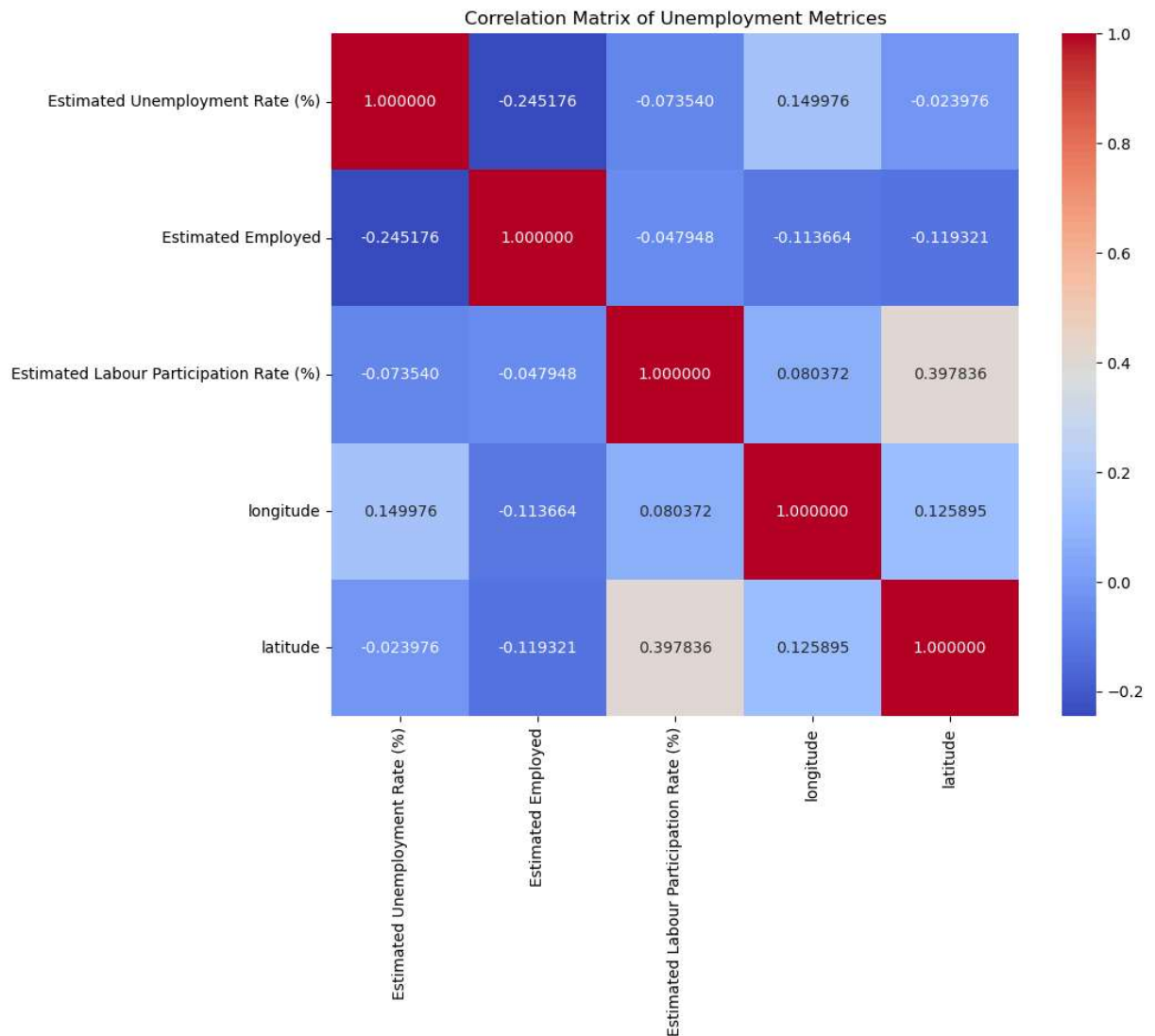


Correlation Analysis

```
In [17]: # Correlation Heatmap with numeric_only=True
plt.figure(figsize=(10,8))

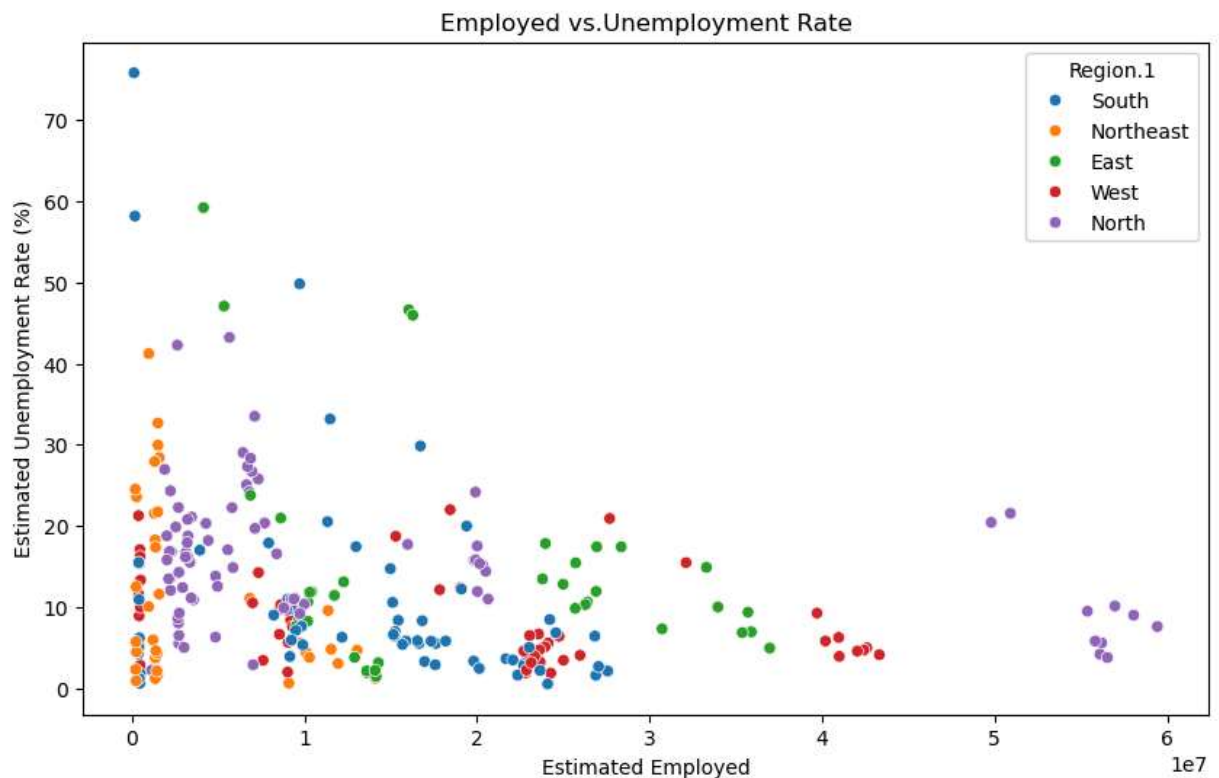
# Only calculate correlation for numeric columns
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm', fmt="2f")
```

```
plt.title('Correlation Matrix of Unemployment Metrics')
plt.show()
```



Comparing Employment vs.Unemployment

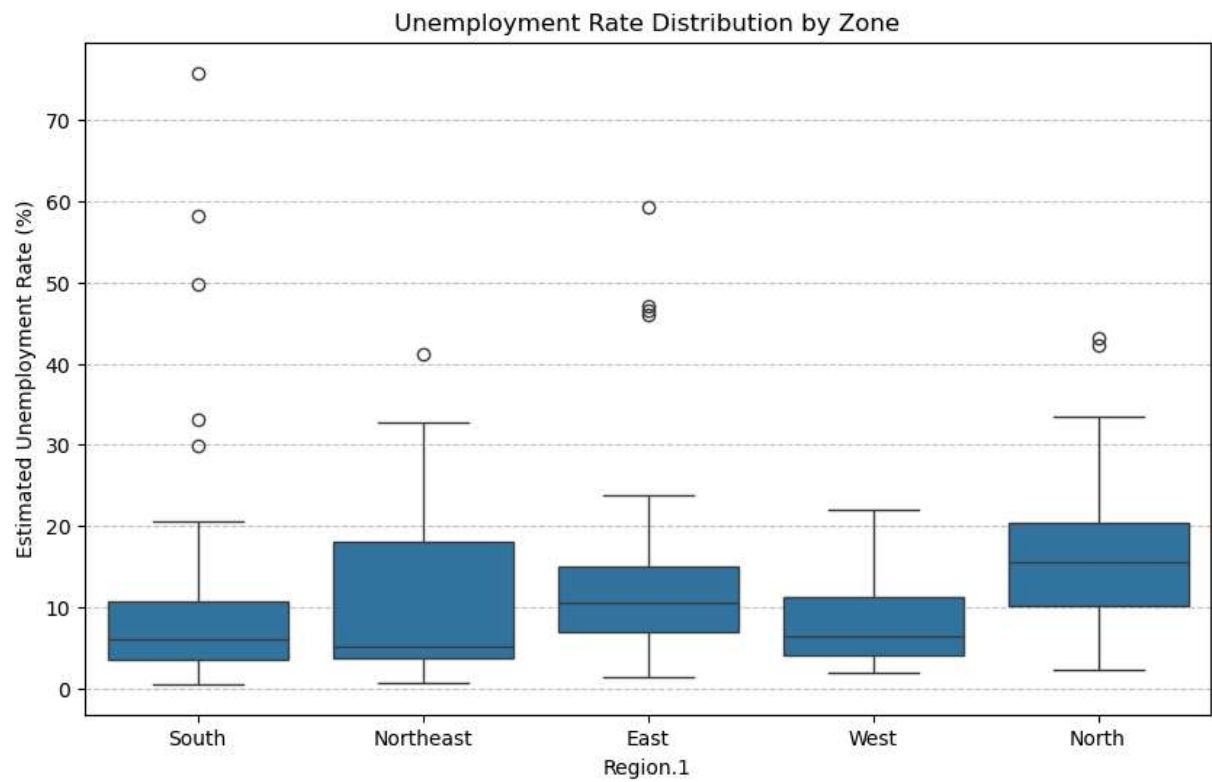
```
In [18]: plt.figure(figsize=(10,6))
sns.scatterplot(data=df, x='Estimated Employed', y='Estimated Unemployment Rate (%)')
plt.title('Employed vs.Unemployment Rate')
plt.show()
```



Distributon Visualization

```
In [20]: import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(10,6))
sns.boxplot(x='Region.1', y='Estimated Unemployment Rate (%)', data=df)
plt.title('Unemployment Rate Distribution by Zone')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```

Thank You